Course Name: Microprocessor and Microcontroller
[Credits: 3; Lecture Hours: 45]

Unit-I [11 Lecture Hours]
Introduction to Microprocessors: History of Microprocessors, Applications, Speed, Word Size, Memory Capacity, Classification of Microprocessors (Mention of Different Microprocessors being used), Microcomputer, Basic Block Diagram of Microcomputer, CPU, ALU, Control Unit, Buses, Memory, Input/Output.
8085 Microprocessor: Features, Architecture, Functional Block Diagram, General Purpose Registers, Register Pairs, Flags, Special Purpose Resisters, Stack Pointer, Program Counter, Types of Buses, Multiplexed Address Bus and Data Bus, Generation of Control Signals, Pin-out Diagram and Pin Description, Basic Memory and I/O Interfacing Concepts, Memory Mapped I/O and I/O Mapped I/O, Partial and Full Address Decoding.

Unit-II [14 Lecture Hours]
8085 Instructions and Programming: Operation Codes, Operands and Mnemonics, Instruction Set, Addressing Modes, Instruction Format, Instruction Classification, Data Transfer Instructions, Arithmetic Instructions, Increment and Decrement Instructions, Logical Instructions, Branch Instructions and Machine Control Instructions, Stack Operations, PUSH and PULL Instructions, Subroutine, Call and Return Instructions, Program Status Word (PSW), Delay Loops, Use of Counters, Instruction Cycle, Machine Cycle, Timing Diagrams, T-States, Op-Code Fetch, Memory Read, Memory Write, I/O Read and I/O Write, Examples of Timing Diagrams for Selective Instructions MVI and STA,Assembly Language Programming and Examples.
8085 Interrupts and Interfacing: Interrupt Structure, EI and DI Instructions, INTR and $\overline{\text{INTA}}$ Signals, Handling Multiple Interrupts using Priority Encoder, RST Instructions, Maskable and Non-Maskable Interrupts, Vectored Interrupts (TRAP, RST 7.5, 6.5 and 5.5) Non-Vectored Interrupts, SIM and RIM Instructions, Latency Time and Response Time, Direct Memory Access (DMA), HOLD and HLDA Signals, Interfacing with Analog to Digital Converter (ADC) IC 0801/0808, Interfacing using General Purpose Programmable Peripheral Interface (PPI)IC 8255, Block Diagram, Pins, Control Word, Different Modes of Operation, BSR and I/O Modes, Illustration of Application of 'Mode 0'.

Unit-III [10 Lecture Hours]
Introduction to Microcontroller: Definition and Use of Microcontroller, Difference with Microprocessor,Types of Microcontrollers, Embedded Microcontrollers, Microcontroller Architectures, Internal Block Diagram, Key Features, Pin Diagram, CISC vs. RISC Architectures, Microcontroller Memory Types, Microcontroller Features, Clocking, I/O Pins, Interrupts, Timers, Peripherals.
8051 Programming: 8051 Addressing Modes and Accessing Memory Locations using Various Addressing Modes, Assembly Language Instructions using Addressing Mode, Arithmetic and Logic Instructions, 8051 Assembly Language Programming Examples, I/O Port Programming in 8051 (using Assembly Language).

Unit-IV [10 Lecture Hours]
Introduction to Arduino Uno: Functional Block Diagram of Arduino Uno Board, ATmega328P: Microcontroller inside Arduino Uno, Functions of different Pins of Arduino Uno, Other Arduino Development Boards, Arduino IDE, Serial Monitor , Arduino Sketch, Variables, setup Function, I/O Functions: pinMode(), digitalRead(), digitalWrite(), analogRead(), analogWrite(), delay() Function, Looping Techniques, Decision Making Techniques, Designing of 1st Sketch, Arduino In-System-Programmer (ISP), Serial Port Communication with PC, Basic Interfacing and I/O Concept, Interfacing LED and Switch, Interfacing with Temperature Sensor LM35.

ELT-H-CC-3-3-P
Course Name: Microprocessor and Microcontroller Lab

[Credit: 1; Lecture Hours: 30]

Section-A: 8085 Microprocessor Assembly Language Programming

1. Program to Transfer a Block of Data.
2. Program for Addition and Subtraction of Numbers using Direct Addressing Mode.
1. Program for Addition and Subtraction of Numbers using Indirect Addressing Mode.
3. Program to Multiply Two 8-Bit Numbersusing Repeated Addition Algorithm.
4. Program to Divide Two 8-Bit Numbersusing Repeated Subtraction Algorithm.
5. Program to Divide a 16-Bit Number by 8-Bit Number.
6. Program to Search a given Number in a given List.
7. Program to Find Number of Zeroes and Ones in an 8-Bit Number.
8. Program to Find Whether an 8-Bit Number is Even or Odd.
9. Use of CALL and RETURN Instruction.
10. Program to Implement a Subroutine Operation.
11. Program to Implement a Stack Operation.
12. Program to Generate Fibonacci Series.
13. Program to Find Minimum and Maximum among N Numbers.
14. Program to Find Square Root of an Integer.
15. Program to Find GCD of Two Numbers.
16. Program to Sort Numbers in Ascending/Descending Order.
17. Program to Verify the Truth Table of Logic Gates.

Section-B: 8051 Microcontroller Assembly Language and Arduino Programming

1. Program to Find Whether the Numbers are Prime or Not.
2. Program to Find Factorial of a Number.
3. Program to Find (a) Largest of N Numbers and (b) Smallest of N Numbers.
4. Program to Arrange Numbers in Ascending/Descending Order.
5. Write and Execute a Sketch in Arduino Uno IDE to Blink an LED.
6. Write and Execute a Sketch in Arduino Uno IDE to Interface a Switch to Turn an LED On and Off.
7. Write and Execute a Sketch in Arduino Uno IDE to Interface with Ttemperature Controller IC LM35 to Monitor Temperature.

Reference Books:

- Gaonkar, Microprocessor Architecture, Programming and Applications with the 8085, Penram.
- B. Ram, Fundamentals of Microprocessors and Microcomputers, Dhanpat Rai.
- Mazidi, McKinlay, Mazidi and Das, Microprocessors and Microcontrollers, Pearson.
- Mathur and Panda, Microprocessors and Microcontrollers, PHI.
- Krishna Kant, Microprocessors and Microcontrollers: Architecture, Programming and System Design, PHI.
- Kumar, Saravanan, and Jeevananthan, Microprocessors and Microcontrollers, Oxford.
- Shah, 8051 Microcontrollers: MCS 51 Family and its Variants, Oxford.
- Ashwin Pajankar,ARDUINO Made Simple, BPB Publications.

CC-4: Mathematical Foundation, Numerical Analysis and Scilab
[Credits: 4 (3TH+1P)]

ELT-H-CC-4-3-TH
Course Name: Mathematical Foundation, Numerical Analysis and Scilab
[Credits: 3; Lecture Hours: 45]

UNIT-I [8 LectureHours]
Vector Analysis: Scalar and Vector Fields, Gradient of a Scalar, Divergence and Curl of a Vector Field, del and Laplacian Operators, Gauss' Divergence, Stokes and Green's Theorems with Simple Applications.
Matrices: Eigenvalues and Eigenvectors, Solving System of Linear Algebraic Equations, Gaussian Elimination Method.

UNIT-II [15LectureHours]
Ordinary Differential Equations: First Order Ordinary Differential Equations, Separable Ordinary Differential Equations, Exact Ordinary Differential Equations and Integrating Factors, Linear Ordinary Differential Equations, Second Order Linear Homogeneous and Inhomogeneous Differential Equations with Constant and Variable Coefficients, Series Solution of Differential Equations, Frobenius Method.
Partial Differentiation Equations: Functions of Several Variables: Limit and Continuity, Partial Differentiation, Variable Treated as Constant, Total Derivative, Partial Differentiation of Composite Functions: Change of Variables, Solution of PDE by Method of Separation of Variables.

Unit-III: [15LectureHours]
Approximations and Errors in Numerical Analysis: Fixed and Floating-Point Representation, Round-Off Errors, Truncation Errors, Absolute and Relative Errors, Error Propagation.
Finding a Root of an Algebraic or Transcendental Equation: Bisection and Newton-Raphson Methods.
Interpolation and Curve-Fitting: Lagrange Interpolation, Newton's Divided Difference Interpolation, Newton's Forward and Backward Interpolation Formula, Divide Difference Tables, Linear Least Squares Fitting.
Numerical Differentiation and Integration: Forward, Backward and Central Difference Formula for Differentiation, Difference Tables, Trapezoidal Rule and Simpson's Composite 1/3 Formula for Numerical Differentiation, Numerical Solution of Ordinary Differential Equation: Euler Method (First Order), Runge Kutta Method(SecondOrder).

Unit-IV: [7LectureHours]
Scilab Fundamentals: Scilab Environment and Console, SciNotes, Scilab Script, Variables, Functions, Vectors and Matrices, Loops, 2D and 3DPlotting.

ELT-H-CC-4-3-P
Course Name: Mathematical Foundation, Numerical Analysis and Scilab Lab
[Credit: 1; Contact Hours: 30]

Implementation with SciLab/MATLAB/Any Other Simulation Software
1. To SolveFirst-Order Ordinary Differential Equation with given Initial Values.
2. To SolveSecond-Order Ordinary Differential Equation with given Initial Values.
3. To Solve an Algebraic or Transcendental Non-Linear Equation using Bisection Method.
4. To Solve an Algebraic or Transcendental Non-Linear Equation using Newton-Raphson's Method.
5. To Implement Lagrange's Interpolation.
6. To Implement Newton's Divided Difference Interpolation.
7. To Integrate a given Function $f(x)$ within limits x_1 to x_2 using Simpson's Composite 1/3 Method.
8. To Find the Derivatives of a Function given in the form of x_i–$f(x_i)$ Tabulated Data using Difference Formulae.
9. To Solve a given First-Order Ordinary Differential Equation using Euler Method.
10. To Solve a System of Linear Algebraic Equations using Gauss Elimination.